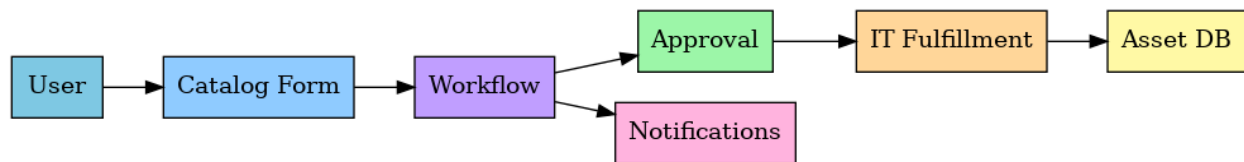


Project Design Phase- II

Technological Stack(Architecture and Stack)

Date	02 NOV 2025
Team ID	NM2025TMID06137
Project Name	Laptop Request Catalog Item
Maximum marks	4 Marks

Technological Architecture:



The architecture for the Laptop Request Catalog system in ServiceNow is designed to simplify and automate the end-to-end laptop provisioning process for employees. The flow starts with the end user, who submits a request through the Service Catalog form in the self-service portal. This form collects key details like laptop type, purpose, and justification.

Once the request is submitted, it passes into the Workflow Engine, which orchestrates the entire approval and fulfillment process. The workflow triggers business rules and scripts that validate input, route the request to the appropriate manager for approval, and apply organization-specific policies. If approved, the request is automatically assigned to the IT Fulfillment team, who are responsible for preparing and issuing the laptop.

Throughout the process, the CMDB (Configuration Management Database) and asset management tables are used to store and update asset records. When IT allocates the laptop, asset assignment links the device to the requester, ensuring accurate inventory and audit tracking. In parallel, notifications keep the user, manager, and IT staff informed at each stage using email and system messages.

This architecture ensures an efficient, transparent, and policy-driven laptop issuance process with automation, role-based access, traceability, and audit compliance.

Components and Technology:

Category	Component / Module	Technology / Tool Used	Purpose
Front-End	Service Catalog Form	ServiceNow Service Catalog	Collect laptop request details from users
Workflow Automation	Approval & Fulfillment Workflow	ServiceNow Flow Designer / Workflow Editor	Automate approval and IT fulfillment process
Backend	Business Rules / Script Includes	ServiceNow Scripting (JavaScript)	Logic execution, request validation, automation
Notification System	Email & System Notifications	ServiceNow Notification Engine	Notify users during each process phase
Database Layer	CMDB & Asset Table	ServiceNow CMDB	Store laptop asset records & assign assets
Access Control	Role Management	ServiceNow ACL & Roles (Employee, Manager, IT)	Ensure secure and role-based access
Reporting	SLA & Status Reporting	SN Reporting / Dashboard	Track SLAs, request status, audit logs
Testing & Deployment	UAT & Promotion	ServiceNow Update Sets / Change Requests	Move changes to production & validate system

Application Characteristics:

Characteristic	Description	Relevance to Project
User Accessibility	Application is accessible via ServiceNow portal	Enables employees to request laptops easily from any device
Usability	Simple, intuitive request submission and status tracking	Reduces user effort and training need
Automation	Automated approval, assignment, and notifications	Eliminates manual processing, reduces delays
Reliability	Ensures consistent performance & request routing	Maintains trust in the system & avoids workflow failures
Maintainability	Workflow and form fields can be updated easily	Supports future enhancements and organizational policy changes
Scalability	Can support growing number of users and requests	Suitable for enterprise environments
Security	Role-based access and audit logging	Protects sensitive data and supports compliance
Performance	Fast load and processing time	Smooth user experience even with large user base
Compliance	Follows IT asset management policies & audit trails	Ensures trackability and accountability
Interoperability	Works with CMDB, Asset Management, Email services	Integrates core ServiceNow modules for seamless flow